

输入信息一：材料物性数据

材料类型	材料ID	HOMO (eV)	LUMO (eV)	Tg (°C)	S1 (eV)	T1 (eV)	PL Peak (nm)	激发态寿命 (t1/t2)	溶解性 (溶剂: 苯甲酸甲酯)	空穴迁移率	电子迁移率	分子化学结构
P-host	SGH5549	-5.33	-2.13	120	3.1	2.65	404/426	5.6ns/-	20mg/mL	SGH5549>P1230>1206	/	
N-host	SGH5516	-5.74	-2.84	125	2.8	2.58	445	5.6ns/-	20mg/mL		SGH5516>N911	
P-host	P1206	-5.25	/	112	3.4	2.74	405/426	4.6ns/-	>80 mg/mL		/	
P-host	P1230	-5.31	/	121	3.39	2.74			30 mg/mL		/	
N-host	N911	-5.71	/	125	2.8	2.72	473	4.8ns/-	>80 mg/mL	/	N911>PH120	
N-host	N1095	-5.69	/	118	2.89	2.81			>80 mg/mL	/		
N-host	N981	-5.65	-2.28	121	2.8	2.74			>80 mg/mL	/		
P-host	PH119	-5.30	-5.30	110	/		407/429	5.3 ns/-	40 mg/mL	/	/	
N-host	PH120	-5.64	-2.34	115	/	2.75	507	22.3ns/1.67us	30 mg/mL	PH120>SGH5516	PH122>PH120	
N-host	PH122	-5.70	-2.40	131	/	2.74	511	23.6ns/5.55us	40 mg/mL			
N-host	SGH531	-5.75	-2.86	130	2.59	2.57	472	7.1ns/-	30 mg/mL	/	SGH5549>SGH531	
磷光敏化剂	PGD508	-5.21	-2.47	/	/	2.34	533	1.5us	10 mg/mL	/	/	
荧光 dopoant	FGD5563	-5.36	/	/	/		535	0.1us	3 mg/mL	/	/	

输入信息二：实验设计&结果

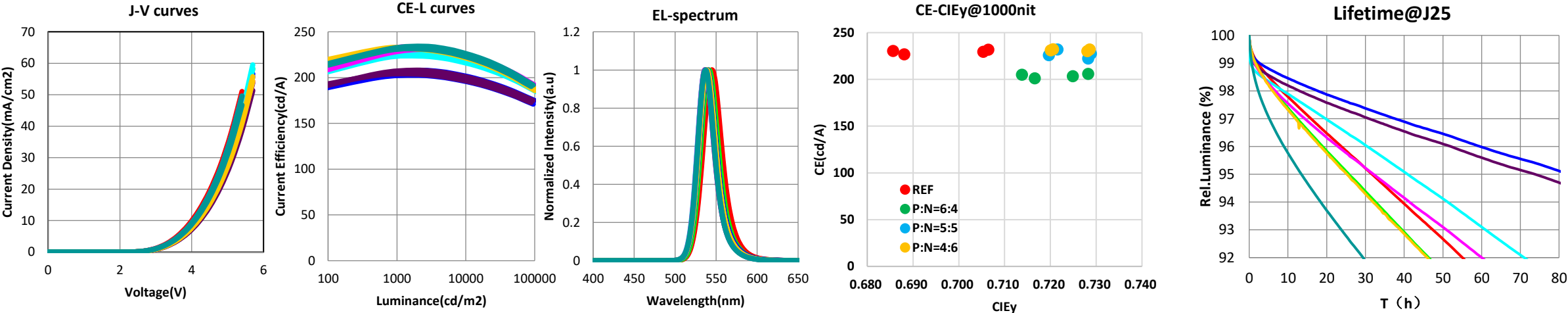


[SPGD508_SFGD5563] PN比例验证 (1225)

实验目的：基于SPGD508+SFGD5563，PN比例改善电平衡；

实验结果：REF(SFGD5501)器件CE~230cd/A@Gy~0.70，EQE~50%，J10电压~4.12V，
SGH5549:SGH5519=6:4/5:5/4:6，Gy~0.72下器件CE分别~205/230/232cd/A，J10电压分别~4.20/4.12/4.08V，

实验结论：SGH5549:SGH5519=6:4 / 5:5搭配PGD508分别作为长寿命、高效率Y4重点方案



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95
	15/110/75nm	PFA147=2:1 120℃ pre 15s 230℃ 30min	PFB258B=1:5 230℃ 30min N2	VD 5min 140℃ 20min	ETL:LiQ	Yb/Ag	CPL008	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25
1	IZO/Ag/IZO UV 2min	1600rpm 30s	1200rpm 30s	SGH5549:SGH5516=4:6 SPGD508:SFGD5501=12%:0.5%	LG1035:LiQ= 4:6 30nm	1/18nm	60nm	4.00	0.299	0.688	0.44	2.73	226.67	48.88	544	26.65	229.8	49.5	3.03	~39
2		1700rpm 30s						4.02	0.302	0.686	0.43	2.74	230.35	49.60	545	26.68	231.1	49.8	3.04	
3		1700rpm 30s		SGH5549:SGH5516=6:4 SPGD508:SFGD5563=12%:0.7%				4.12	0.277	0.706	0.43	2.76	231.53	50.10	541	25.15	232.1	50.2	3.09	~33
4		1800rpm 30s						4.11	0.278	0.705	0.44	2.76	229.29	49.68	541	25.73	230.5	49.9	3.08	
5		1700rpm 30s		SGH5549:SGH5516=5:5 SPGD508:SFGD5563=12%:0.7%				4.14	0.267	0.714	0.49	2.80	204.90	44.72	539	24.78	207.3	45.2	3.13	~110
6		1800rpm 30s						4.16	0.264	0.717	0.50	2.81	201.12	44.07	539	24.97	201.7	44.2	3.15	
7		1700rpm 30s		SGH5549:SGH5516=4:6 SPGD508:SFGD5563=12%:0.7%				4.22	0.253	0.725	0.49	2.83	203.20	44.74	537	24.09	202.7	44.6	3.19	~104
8		1700rpm 30s						4.12	0.249	0.728	0.45	2.77	222.13	48.97	537	24.23	222.8	49.1	3.10	
9		1800rpm 30s						4.11	0.260	0.720	0.44	2.76	225.67	49.31	539	24.34	225.4	49.3	3.09	~52
10		1800rpm 30s						4.04	0.249	0.729	0.44	2.75	227.28	50.03	537	23.86	230.5	50.8	3.05	
11		1700rpm 30s		SGH5549:SGH5516=4:6 SPGD508:SFGD5563=12%:0.7%				4.12	0.258	0.722	0.43	2.76	231.77	50.76	538	24.48	232.4	50.9	3.08	~44
12		1800rpm 30s						4.18	0.260	0.720	0.43	2.77	230.99	50.46	539	24.45	229.8	50.2	3.12	
13		1700rpm 30s						4.17	0.259	0.721	0.43	2.77	232.10	50.85	538	24.84	230.3	50.5	3.11	~33
14		1800rpm 30s						4.08	0.250	0.728	0.43	2.74	230.00	50.52	537	23.83	230.4	50.6	3.07	
15		1800rpm 30s		4.04				0.249	0.729	0.43	2.73	231.95	50.98	537	23.74	233.7	51.4	3.05	~25	

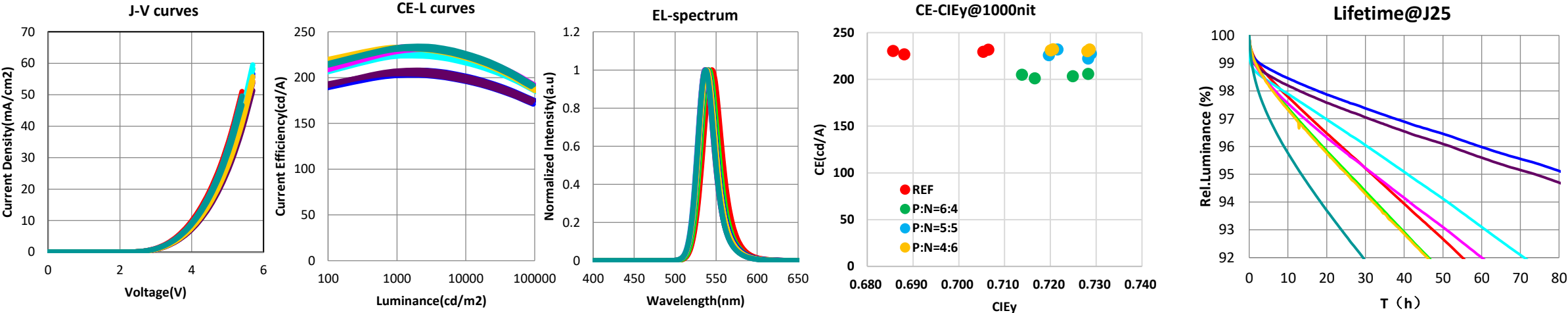


[SPGD508_SFGD5563] PN比例验证 (1225)

实验目的：基于SPGD508+SFGD5563，PN比例改善电平衡；

实验结果：REF(SFGD5501)器件CE~230cd/A@Gy~0.70，EQE~50%，J10电压~4.12V，
SGH5549:SGH5519=6:4/5:5/4:6，Gy~0.72下器件CE分别~205/230/232cd/A，J10电压分别~4.20/4.12/4.08V，

实验结论：SGH5549:SGH5519=6:4 / 5:5搭配PGD508分别作为长寿命、高效率Y4重点方案

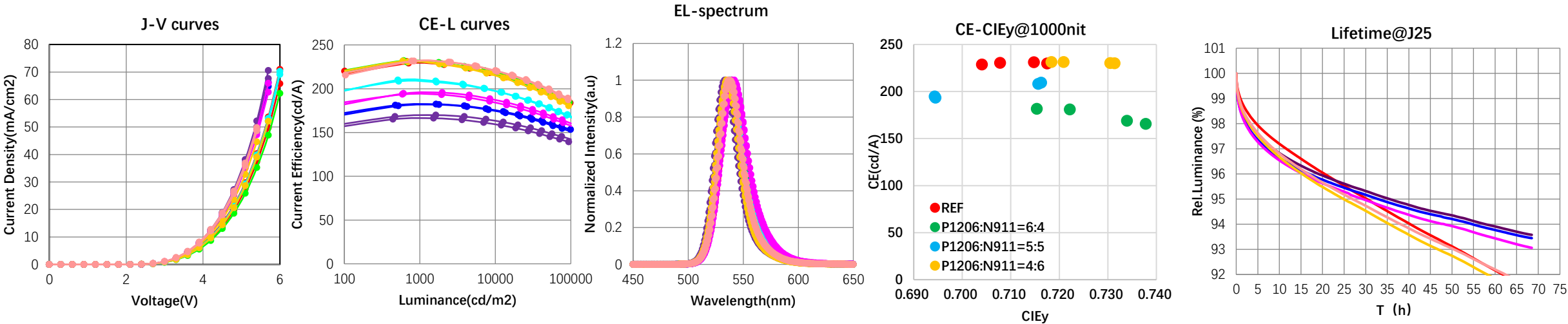


Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95
	15/110/75nm	PFA147=2:1 120℃ pre 15s 230℃ 30min	PFB258B=1:5 230℃ 30min N2	VD 5min 140℃ 20min	ETL:LiQ	Yb/Ag	CPL008	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25
1	IZO/Ag/IZO UV 2min	1600rpm 30s	1200rpm 30s	SGH5549:SGH5516=4:6 SPGD508:SFGD5501=12%:0.5%	LG1035:LiQ= 4:6 30nm	1/18nm	60nm	4.00	0.299	0.688	0.44	2.73	226.67	48.88	544	26.65	229.8	49.5	3.03	~39
2		1700rpm 30s						4.02	0.302	0.686	0.43	2.74	230.35	49.60	545	26.68	231.1	49.8	3.04	
3		1700rpm 30s		SGH5549:SGH5516=6:4 SPGD508:SFGD5563=12%:0.7%				4.12	0.277	0.706	0.43	2.76	231.53	50.10	541	25.15	232.1	50.2	3.09	~33
4		1800rpm 30s						4.11	0.278	0.705	0.44	2.76	229.29	49.68	541	25.73	230.5	49.9	3.08	
5		1700rpm 30s		SGH5549:SGH5516=5:5 SPGD508:SFGD5563=12%:0.7%				4.14	0.267	0.714	0.49	2.80	204.90	44.72	539	24.78	207.3	45.2	3.13	~110
6		1800rpm 30s						4.16	0.264	0.717	0.50	2.81	201.12	44.07	539	24.97	201.7	44.2	3.15	
7		1700rpm 30s		SGH5549:SGH5516=4:6 SPGD508:SFGD5563=12%:0.7%				4.22	0.253	0.725	0.49	2.83	203.20	44.74	537	24.09	202.7	44.6	3.19	~104
8		1700rpm 30s						4.12	0.249	0.728	0.45	2.77	222.13	48.97	537	24.23	222.8	49.1	3.10	
9		1800rpm 30s						4.11	0.260	0.720	0.44	2.76	225.67	49.31	539	24.34	225.4	49.3	3.09	~52
10		1800rpm 30s						4.04	0.249	0.729	0.44	2.75	227.28	50.03	537	23.86	230.5	50.8	3.05	
11		1700rpm 30s		SGH5549:SGH5516=4:6 SPGD508:SFGD5563=12%:0.7%				4.12	0.258	0.722	0.43	2.76	231.77	50.76	538	24.48	232.4	50.9	3.08	~44
12		1800rpm 30s						4.18	0.260	0.720	0.43	2.77	230.99	50.46	539	24.45	229.8	50.2	3.12	
13		1700rpm 30s						4.17	0.259	0.721	0.43	2.77	232.10	50.85	538	24.84	230.3	50.5	3.11	~33
14		1800rpm 30s						4.08	0.250	0.728	0.43	2.74	230.00	50.52	537	23.83	230.4	50.6	3.07	
15		1800rpm 30s		4.04				0.249	0.729	0.43	2.73	231.95	50.98	537	23.74	233.7	51.4	3.05	~25	

[莱特2nd GH] SPGD508-PN比例验证(0310)



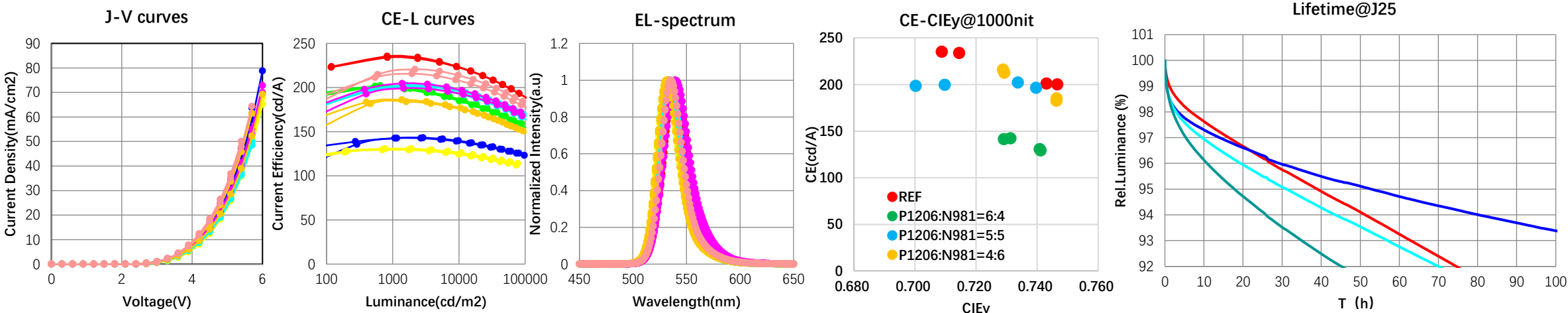
实验目的：SPGD508+SFGD5563，验证莱特GH性能；
实验结果：REF(SGH5549+SGH5516)器件CE~230cd/A@Gy~0.72，J10电压~4.2V，J25实测寿命~；
P1206:N911=6:4/5:5/4:6，器件效率分别~180/208/230cd/A@Gy~0.72，J10电压分别~4.08V/4.06/4.05V；



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95
	15/110/75nm	PFA147=2:1 120℃ pre 15s 230℃ 30min	PFB258B=1:5 230℃ 30min N2	VD 5min 140℃ 20min	ETL:LiQ	Yb/Ag	CPL008	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25
1	IZO/Ag/IZO UV 2min	1700rpm 30s	1200rpm 30s	SGH5549:SGH5516=5:5 SPGD508:SFGD5563=12%:0.7%	LG1035:LiQ= 4:6 30nm	1/18nm	60nm	4.24	0.274	0.708	0.44	2.78	230.37	50.34	540	26.50	226.5	49.5	3.14	52
2		1800rpm 30s						4.18	0.279	0.704	0.44	2.76	228.69	49.93	540	27.07	226.0	49.3	3.11	
								4.29	0.266	0.715	0.43	2.80	231.16	50.67	539	25.79	227.2	49.8	3.17	45
3		1700rpm 30s						4.21	0.262	0.718	0.44	2.77	229.81	50.46	538	25.53	226.4	49.7	3.13	
				4				1800rpm 30s	P1206:N911=6:4 SPGD508:SFGD5563=12%:0.7%	4.08	0.265	0.715	0.55	2.84	181.57	39.80	539	24.63	181.0	39.7
4.03		0.256								0.722	0.55	2.81	180.82	39.93	537	24.59	180.6	39.9	3.15	
5		1700rpm 30s		4.06				0.234		0.738	0.60	2.85	165.65	37.24	534	24.09	164.9	37.1	3.21	85
				4.03				0.240		0.734	0.59	2.84	168.88	37.77	534	24.20	168.8	37.7	3.18	
6		1800rpm 30s		P1206:N911=5:5 SPGD508:SFGD5563=12%:0.7%				4.22	0.263	0.716	0.48	2.83	209.29	46.17	537	25.75	206.3	45.5	3.20	75
								4.21	0.264	0.716	0.48	2.82	207.98	45.73	538	25.30	205.4	45.1	3.19	
7	1700rpm 30s	4.05	0.289					0.694	0.52	2.79	193.36	42.37	541	28.59	191.7	42.0	3.13			
		4.06	0.290					0.694	0.51	2.79	194.66	42.56	542	28.24	194.2	42.5	3.12			
8	1800rpm 30s	P1206:N911=4:6 SPGD508:SFGD5563=12%:0.7%	4.22	0.245				0.731	0.44	2.80	230.29	51.12	536	24.42	225.6	50.1	3.16	50		
			4.12	0.245				0.731	0.44	2.76	230.04	50.96	536	23.81	226.7	50.2	3.10			
			4.02	0.258				0.721	0.43	2.73	231.31	50.99	537	25.16	229.3	50.6	3.04	55		
			4.05	0.261				0.718	0.43	2.74	231.26	50.76	538	24.98	229.3	50.3	3.06			

实验结果：REF器件效率~235cd/A@Gy~0.71，J10电压~4.15V；寿命LT95@J25~55h

P1206:N981=6:4/5:5/4:6，器件效率分别~142/200 (85%)/217 (94%)cd/A，J10电压分别~4.14V/4.24V/4.06V；寿命LT95@J25~130(235%)/62(113%)/43h(78%)



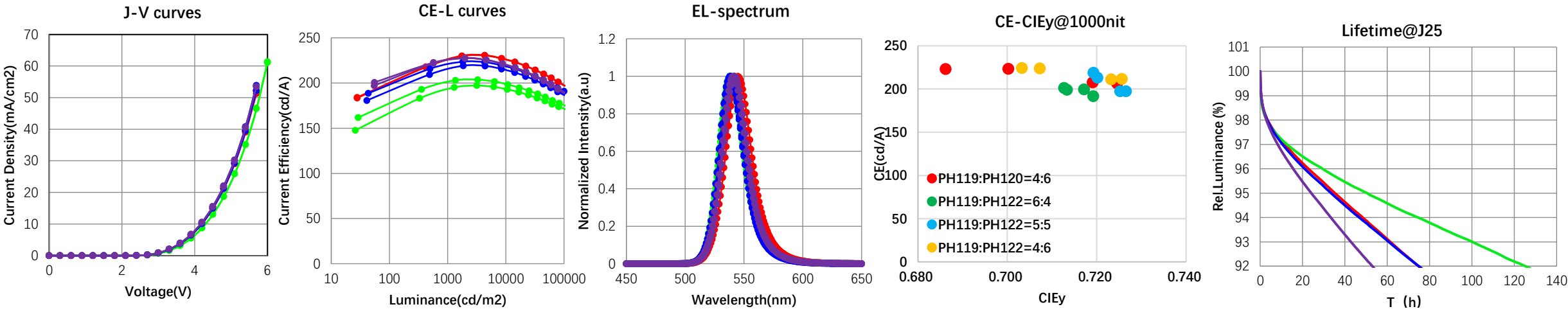
Device	EML	J10	@1000 nit							@3000nit			LT95(AF=1.32)	
	VD 5min 140℃ 20min	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25
1	SGH5549:SGH5516=5:5 SPGD508:SFGD5563=12%:0.7%	4.15	0.266	0.714	0.43	2.73	233.80	51.33	538	25.47	231.7	50.9	3.07	55
		4.15	0.273	0.709	0.43	2.73	235.25	51.54	539	26.21	233.0	51.1	3.07	
2		4.19	0.221	0.747	0.50	2.84	200.04	45.56	532	24.11	194.7	44.3	3.26	/
		4.19	0.227	0.743	0.50	2.79	200.95	45.46	533	24.02	196.4	44.4	3.17	
3	P1206:N981=6:4 SPGD508:SFGD5563=12%:0.7%	4.11	0.246	0.729	0.70	2.95	141.44	31.61	535	23.81	143.1	32.0	3.32	130
		4.16	0.242	0.731	0.70	2.97	142.27	31.93	534	23.99	142.4	32.0	3.34	
4		4.33	0.227	0.741	0.77	3.09	129.31	29.52	531	24.37	128.4	29.3	3.51	62
		4.29	0.228	0.741	0.77	3.06	130.68	29.77	532	23.92	130.0	29.6	3.47	
5	P1206:N981=5:5 SPGD508:SFGD5563=12%:0.7%	4.18	0.271	0.710	0.50	2.84	199.55	43.85	539	26.18	201.7	44.3	3.19	62
		4.21	0.282	0.700	0.50	2.85	198.45	43.52	540	27.76	199.7	43.8	3.21	
6		4.26	0.240	0.734	0.49	2.88	202.09	45.13	535	23.84	203.8	45.5	3.25	43
		4.22	0.232	0.740	0.51	2.87	196.36	44.12	534	23.26	198.6	44.6	3.23	
7	P1206:N981=4:6 SPGD508:SFGD5563=12%:0.7%	4.13	0.221	0.746	0.54	2.88	185.11	42.29	531	24.00	184.3	42.1	3.26	43
		4.11	0.221	0.746	0.55	2.84	182.64	41.66	531	23.60	182.6	41.6	3.19	
8		4.08	0.246	0.729	0.47	2.80	212.71	47.28	535	23.77	214.6	47.7	3.12	43
		4.04	0.247	0.729	0.46	2.78	216.16	48.03	536	23.70	219.9	48.9	3.09	

LT素材- GH_P/N验证

实验目的：LT新款GH验证；

实验结果：REF(PH120)器件CE~223cd/A@Gy~0.70，J10电压~4.2V；J25实测寿命~65h；

PH119:PH120=6:4/5:5/4:6，器件效率分别~200/216/224cd/A，J10电压分别~4.3V/4.2V/4.15V；J25实测寿命分别~120/66/45h；



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95(AF=1.32)	
	15/110/75nm	~50nm	~35nm	~55nm	ETL:LiQ	Yb/Ag	CPL008	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25	@1000nit
1	IZO/Ag/IZO	HIL07	HTL05	PH119:PH120=4:6 SPGD508:SFGD5501=12%:0.5%	LG1035:LiQ=4:6 30nm	1/18nm	60nm	4.19	0.301	0.686	0.44	2.83	222.86	47.98	545	25.55	230.1	49.5	3.14	65	
2				PH119:PH122=6:4 SPGD508:SFGD5501=12%:0.5%				4.20	0.285	0.700	0.44	2.83	223.22	48.26	542	24.94	230.4	49.8	3.15		
3				PH119:PH122=5:5 SPGD508:SFGD5501=12%:0.5%				4.29	0.264	0.717	0.50	2.89	199.39	43.47	539	24.35	203.5	44.4	3.26	120	
4				PH119:PH122=4:6 SPGD508:SFGD5501=12%:0.5%				4.29	0.261	0.719	0.52	2.91	191.57	41.85	539	24.48	197.1	43.1	3.27		
								4.16	0.260	0.720	0.46	2.81	213.06	46.50	539	24.31	219.0	47.8	3.13	66	
								4.19	0.262	0.719	0.45	2.81	218.75	47.56	539	23.48	223.3	48.6	3.14		
								4.16	0.276	0.707	0.44	2.79	223.59	48.44	541	24.72	226.6	49.1	3.12	45	
								4.17	0.281	0.703	0.44	2.79	224.24	48.40	542	24.38	227.0	49.0	3.12		

[SG5516:PH120] 双N比例验证

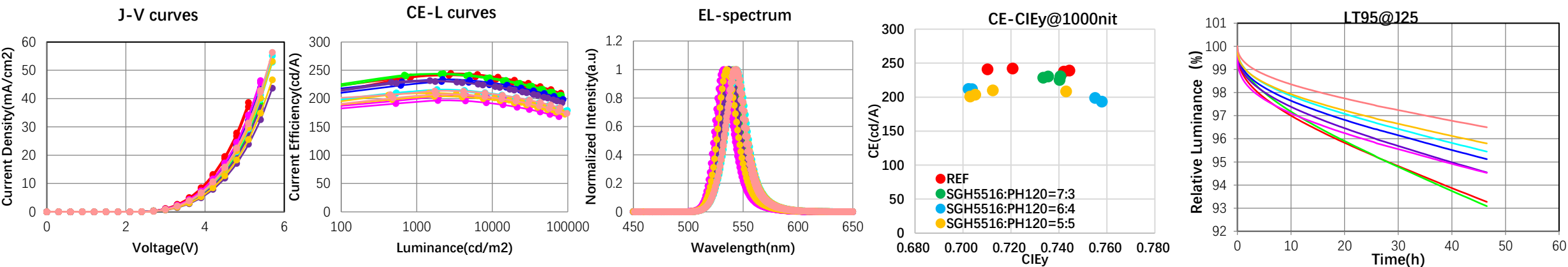


实验目的：双N比例调节验证寿命；

实验结果：基于SPGD508+SFGD5563组合，REF器件效率@3kint~243cd/A，J10电压~4.1V,寿命~52h；

N-host组分为SGH5516:PH120=7:3/6:4/5:5，器件效率分别~230 (-5%)/215(-11%)/210 (-13%)cd/A，J10电压暂无明显差异；寿命分别为78 (+50%)/86 (65%)/106h (+104%)

实验结论：SGH5549:SGH5516:PH120=5:(5:5)搭配PGD508寿命表现优异，后续作为长寿命高潜备案



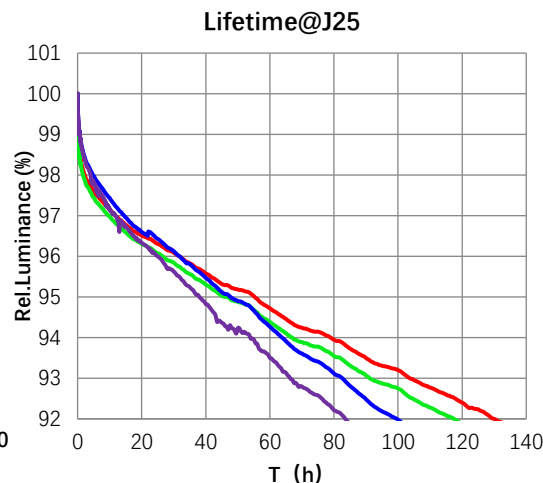
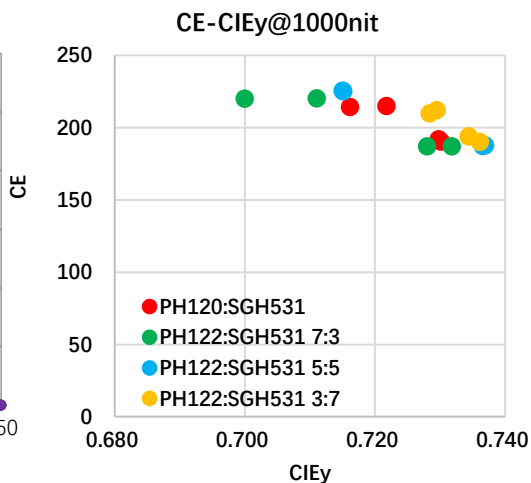
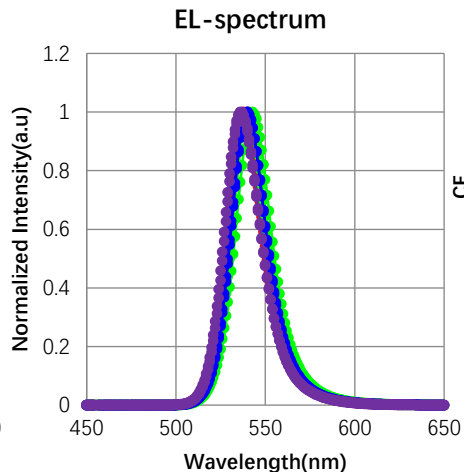
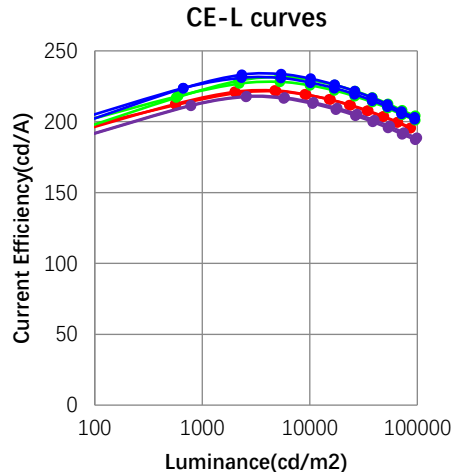
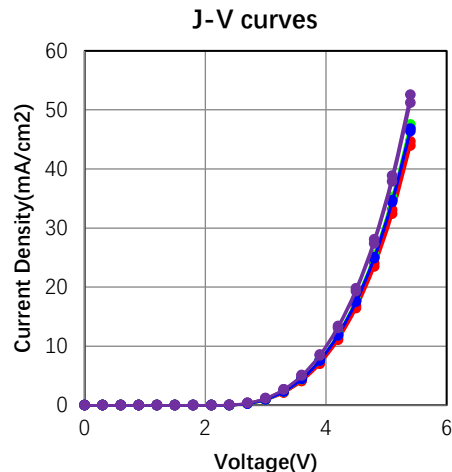
Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95
	15/110/75nm	PFA147=2:1 120℃ pre 15s 230℃ 30min	PFB258B=1:5 230℃ 30min N2	VD 5min 140℃ 20min	ETL:LiQ	Yb/Ag	CPL038	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25
1	IZO/Ag/IZO UV 2min	1700rpm 30s	1200rpm 30s	SGH5549:SGH5516=5:5 SPGD508:SFGD5563=12%:0.7%	LG1035:LiQ= 4:6 30nm	1/18nm	65nm	4.00	0.232	0.744	0.42	2.72	238.79	52.38	536	18.53	244.1	53.6	3.01	54
2								4.02	0.236	0.742	0.42	2.73	237.31	51.77	537	17.71	240.9	52.6	3.03	
3		1800rpm 30s		SGH5549:SGH5516:PH120=5:(7:3) SPGD508:SFGD5563=12%:0.7%				4.23	0.262	0.721	0.41	2.77	241.93	52.20	541	20.14	242.2	52.2	3.11	50
4								4.18	0.275	0.710	0.41	2.75	240.81	51.66	543	20.50	243.3	52.2	3.08	
5		1700rpm 30s		SGH5549:SGH5516:PH120=5:(6:4) SPGD508:SFGD5563=12%:0.7%				4.12	0.237	0.740	0.44	2.78	225.06	49.26	537	18.88	229.8	50.3	3.09	80
6								4.12	0.237	0.741	0.44	2.77	223.92	48.90	537	17.97	226.6	49.5	3.09	
7		1800rpm 30s		SGH5549:SGH5516:PH120=5:(5:5) SPGD508:SFGD5563=12%:0.7%				4.26	0.243	0.735	0.43	2.86	230.29	50.28	537	19.22	229.9	50.2	3.23	75
8								4.26	0.246	0.734	0.44	2.81	228.36	49.64	538	18.83	229.2	49.8	3.17	
		1700rpm 30s		SGH5549:SGH5516:PH120=5:(6:4) SPGD508:SFGD5563=12%:0.7%				4.20	0.283	0.702	0.47	2.81	211.66	45.49	544	22.67	213.3	45.9	3.15	90
								4.18	0.282	0.704	0.47	2.81	211.56	45.37	544	21.15	215.1	46.1	3.14	
		1800rpm 30s		SGH5549:SGH5516:PH120=5:(6:4) SPGD508:SFGD5563=12%:0.7%				4.09	0.211	0.758	0.51	2.80	193.07	43.51	532	19.58	196.4	44.3	3.13	82
								4.09	0.216	0.755	0.50	2.80	198.54	44.32	533	18.67	203.8	45.5	3.12	
		1700rpm 30s		SGH5549:SGH5516:PH120=5:(5:5) SPGD508:SFGD5563=12%:0.7%				4.18	0.283	0.703	0.49	2.83	200.68	43.13	544	21.83	203.4	43.7	3.17	108
								4.16	0.232	0.743	0.48	2.87	208.25	45.96	535	20.50	208.5	46.0	3.25	
		1800rpm 30s		SGH5549:SGH5516:PH120=5:(5:5) SPGD508:SFGD5563=12%:0.7%				4.15	0.280	0.705	0.49	2.82	203.37	43.66	544	21.80	207.0	44.4	3.15	105
								4.16	0.272	0.712	0.47	2.83	209.71	45.06	542	20.19	213.2	45.8	3.15	

LT素材- GH_多元GH比例验证

实验目的: GM2213方案下, PH120替换为PH122评估器件性能;

实验结果: REF(GM2213)器件CE~214cd/A@Gy~0.70, J10电压~4.1V; J25实测寿命预计~120h;

PH122:SGH531=7:3/5:5/3:7, 器件效率分别~220cd/A@Gy~0.70/225cd/A@Gy~0.71/212cd/A@Gy~0.73, J10电压分别~4.07V/4.07V/4.0V; J25实测寿命预计分别~112/85/72h;



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95(AF=1.32)	
	15/110/75nm	~50nm	~35nm	~55nm	ETL:LiQ	Yb/Ag	CPL038	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25	@1000nit
1	IZO/Ag/IZO	HIL07	HTL05	PH119:PH120:SGH531=6:(5:5) SPGD500:SFGD5501=12%:0.5% GM2213	LG1035:LiQ= 4:6 30nm	1/18nm	65nm	4.12	0.259	0.722	0.46	2.79	214.98	46.88	538	23.23	221.3	48.3	3.11	~120	
2				PH119:PH122:SGH531=6:(7:3) SPGD500:SFGD5501=12%:0.5%				4.10	0.266	0.716	0.46	2.79	214.20	46.45	540	22.82	220.6	47.8	3.10		
3				PH119:PH122:SGH531=6:(5:5) SPGD500:SFGD5501=12%:0.5%				4.07	0.286	0.700	0.45	2.78	220.08	47.38	543	23.37	229.5	49.4	3.08	~112	
4				PH119:PH122:SGH531=6:(3:7) SPGD500:SFGD5501=12%:0.5%				4.06	0.273	0.711	0.45	2.77	220.28	47.63	541	22.80	227.2	49.1	3.08		
								4.07	0.267	0.715	0.44	2.76	225.08	48.77	540	22.81	230.8	50.0	3.07	~85	
								4.07	0.267	0.715	0.44	2.76	225.43	48.89	540	23.23	233.2	50.6	3.06		
								4.00	0.248	0.730	0.47	2.74	211.94	46.62	536	23.77	217.3	47.8	3.04	~72	
								3.99	0.250	0.728	0.47	2.74	212.41	46.62	537	23.42	217.9	47.8	3.04		

[SPGD508_SFGD5563] 莱特氙代主体验证 (0509)

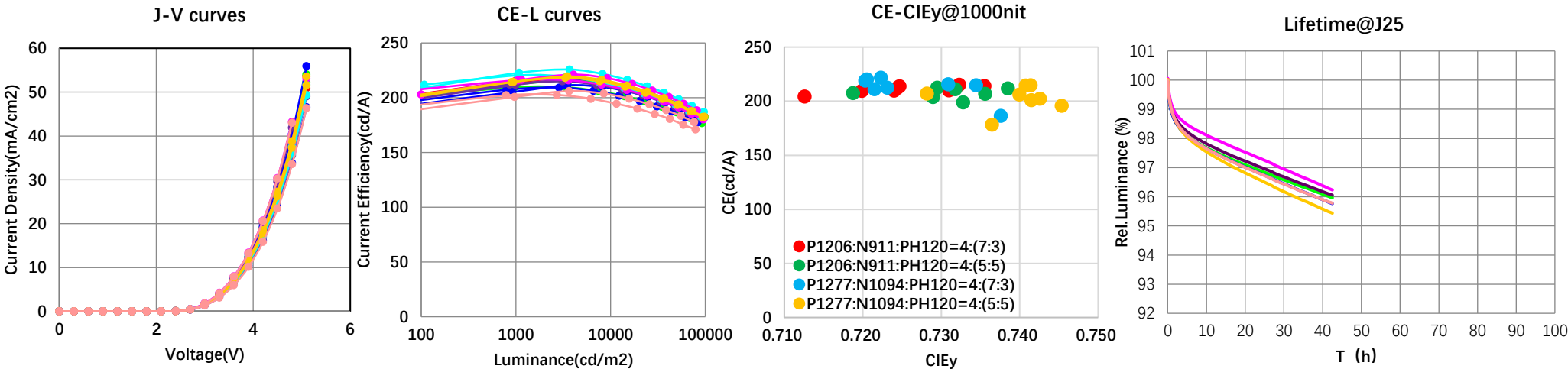


实验目的：莱特氙代主体+PH120电平衡调节验证；

实验结果：P1277=P1206高氙代， N1094=N911高氙代；

P1206:N911=4:6基础下， N911:PH120=7:3/5:5， 器件效率分别~215/213cd/A@Gy~0.72， J10电压分别~3.78V/3.75V； J25实测寿命预计分别~90/89h；

P1277:N1094=4:6基础下， N1094:PH120=7:3/5:5， 器件效率分别~220/215cd/A@Gy~0.74， J10电压分别~3.78V/3.81V； J25实测寿命预计分别~90/89h；



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95(AF=1.32)	
	15/110/75nm	PFA147=2:1 120°C pre 15s 230°C 30min	PFB258B=1:5 230°C 30min N2	VD 5min 140°C 20min	ETL:LiQ	Yb/Ag	CPL038	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25	@1000nit
1	IZO/Ag/IZO UV 2min	1700rpm 30s	1200rpm 30s	P1206:(N911:PH120)=4:(7:3) PGD508:FGD5563=12%:0.7%	LG1035:LiQ= 4:6 30nm	1/18nm	65nm	3.81	0.245	0.732	0.46	2.70	215.08	47.41	536.00	21.54	216.9	47.8	2.95	/	
2		1800rpm 30s						3.78	0.254	0.725	0.47	2.70	213.81	46.92	537.00	22.56	217.2	47.7	2.93		
3		1700rpm 30s		3.76				0.255	0.724	0.47	2.70	212.17	46.47	538.00	21.92	214.9	47.1	2.92	90		
																					3.79
4		1800rpm 30s		3.88				0.262	0.719	0.48	2.73	207.45	45.41	538.00	23.94	209.1	45.8	3.01	87		
																					3.75
5		1700rpm 30s		3.73				0.236	0.738	0.47	2.70	211.78	47.09	535.00	21.59	215.4	47.9	2.92	90		
																					3.84
6		1800rpm 30s		3.79				0.257	0.722	0.45	2.68	221.68	48.52	538.00	22.31	225.0	49.2	2.92	/		
																					3.83
7		1700rpm 30s		3.71				0.242	0.734	0.46	2.66	214.56	47.42	536.00	21.47	219.4	48.5	2.89	90		
																					3.78
8		1800rpm 30s		3.81				0.233	0.741	0.47	2.71	214.48	47.71	534.96	20.88	217.7	48.4	2.96	80		
																					3.78
				3.72				0.232	0.741	0.50	2.70	200.82	44.81	534.00	21.59	204.8	45.7	2.93	89		
																					3.89

[SPGD508_SFGD5563]定版方案验证 (0520)-CIC200-LYK

TCL

实验目的：定版方案验证；

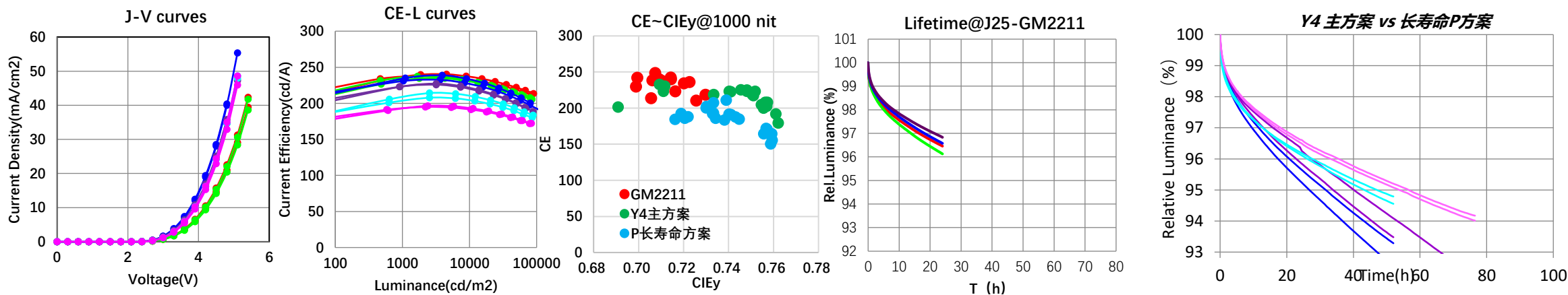
实验结果：GM2211, CE~230cd/A@CIEy~0.70, J10电压~4.22V,

Y4 主方案: P1277:N1094:PH120=4:(7:3), CE~233cd/A@CIEy~0.73, J10电压~3.77V, 寿命LT95@J25~55-65h (偏低20%-30%)

P客户长寿命方案: P1277:N1094:PH120=5:(5:5), CE~210cd/A@CIEy~0.73, J10电压~3.89V, 寿命LT95@J25~110-115h (偏低20%-30%)

【主要结论】相比Y4主方案, P客户长寿命方案效率降低10%, 寿命提升80%, 预估寿命可达成P项目寿命目标的115% (20000h@1knit)

【待确认结论】GM2211与Y4主方案性能差异需要复验明确收益 (Y4主方案色点偏离, 寿命离散且偏低)



Device	Anode	HI	HT	EML	ETL	Cathode	CPL	J10	@1000 nit								@3000nit			LT95(AF=1.32)	
	15/110/75nm	PFA147=2:1 120°C pre 15s 230°C 30min	PFB258B=1:5 230°C 30min N2	VD 5min 140°C 20min	ETL:LiQ	Yb/Ag	CPL038	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25	@1000nit
1	IZO/Ag/IZO UV 2min	1550rpm 30s	1200rpm 30s	GM2211	LG1035:LiQ= 4:6 30nm	1/18nm	65nm	4.16	0.274	0.711	0.42	2.81	234.73	50.35	541.63	21.47	240.0	51.5	3.13	90	
2		1650rpm 30s						4.22	0.275	0.710	0.42	2.83	235.65	50.49	542.00	21.19	237.3	50.8	3.16		
3		1700rpm 30s						4.24	0.262	0.720	0.42	2.83	234.15	50.53	540.00	21.57	237.0	51.1	3.17	85	
								4.19	0.288	0.699	0.43	2.82	229.82	49.27	543.00	24.86	234.5	50.3	3.14		
4		1800rpm 30s		P1277:N1094:PH120=4:(7:3) PGD508:FGD5563=12%:0.7%				3.77	0.272	0.712	0.43	2.69	230.40	49.88	541.00	22.49	232.2	50.3	2.93	55	
								3.75	0.274	0.709	0.43	2.67	232.68	50.33	542.00	22.65	237.6	51.4	2.90		
5		1700rpm 30s		P1277:N1094:PH120=5:(5:5) PGD508:FGD5563=12%:0.7%				3.84	0.219	0.752	0.45	2.71	223.15	50.03	533.00	19.58	225.6	50.6	2.97	65	
								3.85	0.223	0.748	0.45	2.71	223.04	49.90	533.96	21.07	226.9	50.7	2.97		
6		1800rpm 30s						3.88	0.248	0.730	0.49	2.76	202.28	44.32	537.00	20.89	207.7	45.5	3.04	115	
								3.89	0.244	0.733	0.48	2.76	207.30	45.52	537.00	20.42	214.1	47.0	3.04		
	3.88		0.235		0.740	0.52	2.77	191.58	42.40	535.00	20.16	196.4	43.5	3.05	110						
3.92	0.245	0.733	0.52	2.78	192.16	42.20	537.00	20.37	194.8	42.8	3.07										

[SPGD5622_SFGD5563] LT-GH掺杂寿命优化验证(0429)-trovato

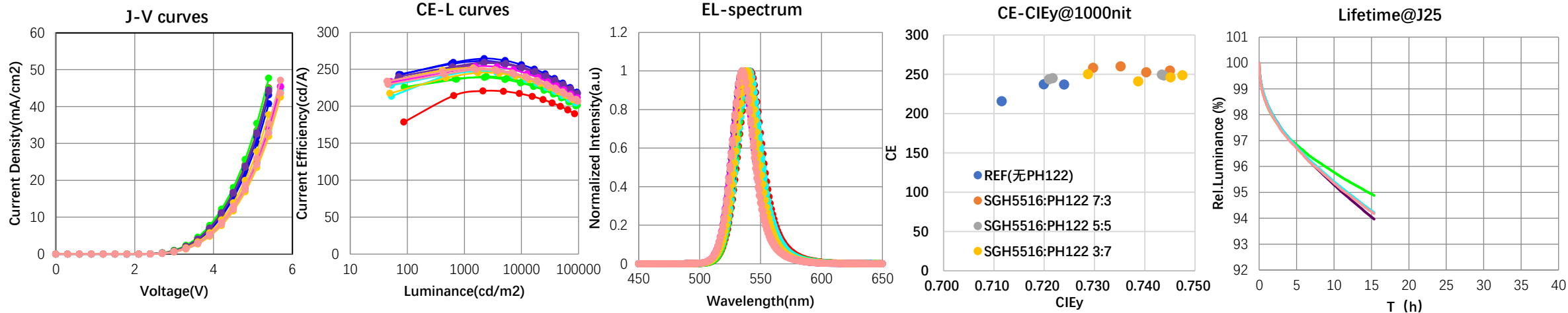
TCL

实验目的：在SPGD5622体系中掺杂PH122，寿命优化验证；

实验结果：SPGD508敏化体系器件效率~237cd/A@Gy~0.72，J10电压~4.1V；J25实测寿命预计~31h；

SPGD5622+SFGD5563敏化体系，SGH5516:PH122=7:3/5:5/3:7，效率分别~260/249/250cd/A@Gy~0.72，J10电压分别~4.12V/4.32V/4.30V；J25实测寿命预计分别~20/22/23h；

初步结论：PGD光谱蓝移对PSF器件效率影响显著，相比PGD508，PGD5622效率提升8%，寿命降低~50%，选型GH以提升PGD5622寿命作为高优先级任务推动



Device	Anode	HI	HT	EML	ET	Cathode	CPL	J10	@1000 nit								@3000nit			LT95(AF=1.32)	
	15/110/75nm	PFA147 230°C 30min	PFB258B=1:5 230°C 30min N2	VD 5min 140°C 20min	ETL:LiQ	Yb/Ag	CPL	V(v)	CIE x	CIE y	J(mA/cm2)	V(v)	C.E.(cd/A)	EQE (%)	WP(nm)	FWHM(nm)	C.E.(cd/A)	EQE (%)	V(v)	@J25	@1000nit
1	IZO/Ag/IZO UV 2min trovato蒸镀	2:1 1500rpm/30s	PFB258B 1200rpm 30s 35nm	SGH5549:SGH5516 5:5 SPGD508:SFGD5563 12%:0.7% 1800rpm/30s	LG1035:LiQ =4:6 30nm	1/18nm	CPL038 65nm	4.09	0.272	0.712	0.46	2.77	215.96	46.68	541	23.14	220.6	47.7	3.09	31	
2		1600rpm/30s						4.05	0.257	0.724	0.42	2.75	237.06	51.52	539	21.44	239.8	52.1	3.05		
3		1500rpm/30s						4.10	0.262	0.720	0.42	2.76	237.45	51.49	540	22.12	238.2	51.7	3.08		
4		1600rpm/30s		SGH5549:SGH5516:PH122=5:(7:3) SPGD5622:SFGD5563 12%:0.7%				4.19	0.249	0.730	0.39	2.78	258.31	56.32	539	21.97	260.5	56.8	3.10	20	
5		1500rpm/30s						4.14	0.242	0.735	0.38	2.77	260.34	57.21	537	22.33	263.8	58.0	3.08		
6		1600rpm/30s						4.11	0.229	0.745	0.39	2.77	254.87	56.41	536	21.24	259.2	57.4	3.07		
7		1500rpm/30s		SGH5549:SGH5516:PH122=5:(5:5) SPGD5622:SFGD5563 12%:0.7%				4.12	0.236	0.740	0.39	2.77	252.67	55.57	537	21.75	257.7	56.7	3.08	22	
8		1600rpm/30s						4.33	0.259	0.722	0.41	2.85	245.10	53.33	540	23.16	246.4	53.6	3.20		
								4.32	0.260	0.721	0.41	2.85	243.98	52.87	541	22.40	246.6	53.4	3.20		
								4.32	0.231	0.743	0.40	2.85	249.86	55.22	536	21.20	253.4	56.0	3.20		
								4.32	0.230	0.744	0.40	2.85	248.66	55.17	535	21.94	249.7	55.4	3.20		
								4.25	0.237	0.739	0.41	2.83	241.22	53.18	536	21.80	244.9	54.0	3.16		
				SGH5549:SGH5516:PH122=5:(3:7) SPGD5622:SFGD5563 12%:0.7%				4.37	0.251	0.729	0.40	2.87	250.44	54.49	539	21.45	249.8	54.3	3.23	23	
								4.34	0.226	0.747	0.40	2.86	248.99	55.25	535	20.66	249.6	55.4	3.22		
								4.30	0.228	0.745	0.40	2.85	246.23	54.67	535	21.46	248.4	55.2	3.19		

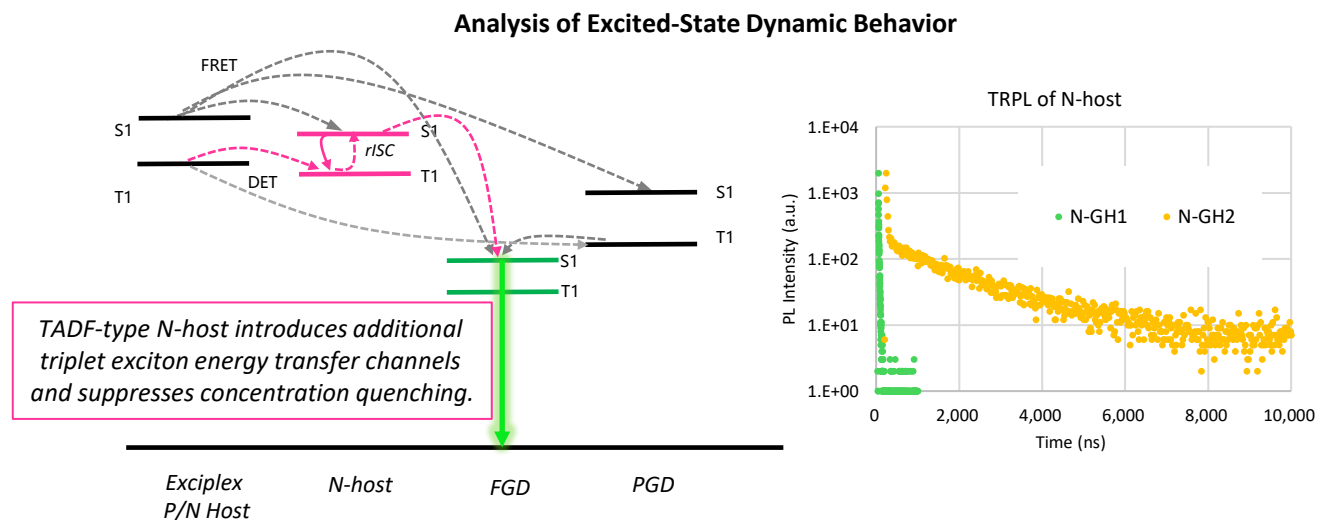
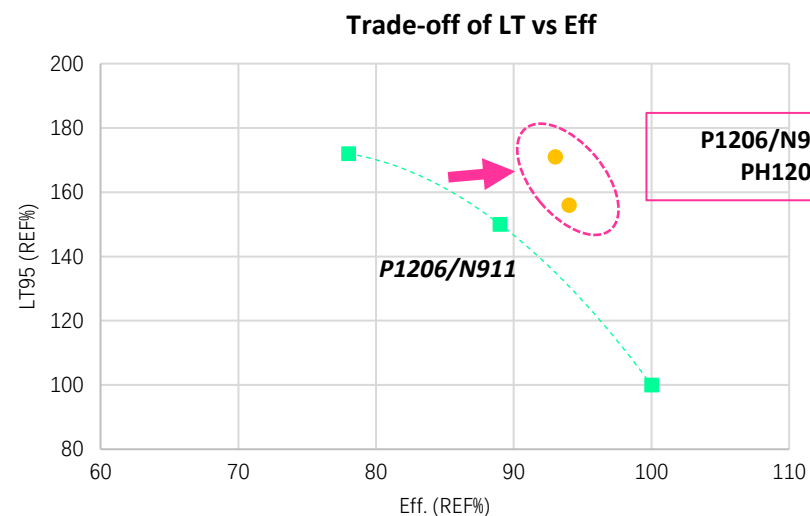
输入信息三：机理分析/经验判断

PSF器件寿命优化：多组分GH共混

【技术背景】PSF器件主要通过双组分GH优化电荷平衡，存在效率-寿命的严重Trade-off

【寿命优化进展】多组分主体共混（P/N-host+增强TADF型-n-host）方案实现电荷平衡的精细化调控与激发态能量转移路径的优化，保证寿命收益同时减缓效率降幅

【后续计划】完善多主体共混的设计原理和材料物性数据库，赋能EL Agent支撑多组分GH方案优化@6/10



ID	P-Host	N-Host1	N-Host2	PGD	FGD	P/N ratio	V@J10 (REF%)	CE (REF%)	LT95 (REF%)
1	P1206	N911	/	PGD508	FGD5563	6:4	100	78	172
2			/			5:5	104	89	150
3			/			4:6	100	100	100
4			PH120			4:(7:3)	100	94	156↑
5			PH120			4:(5:5)	104	93	170↑

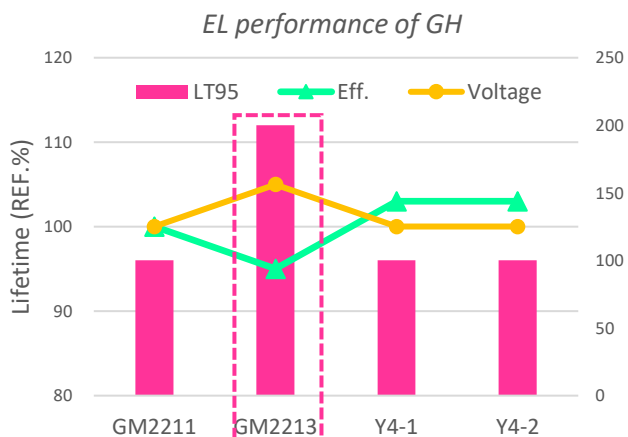
GH对寿命影响分析

TCL

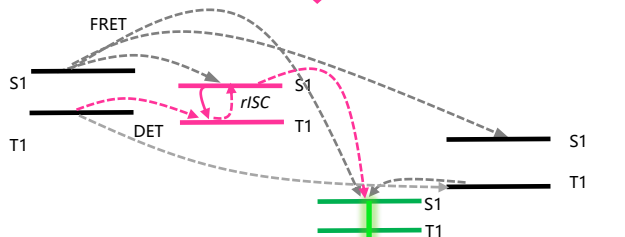
【GH关键影响】LT-GH调控可实现寿命+100%，效率降低~5%

【光物理分析进展】P-host能级/带隙相近，长寿命GH对应N-host表现为强TADF特性（TSCT分子），通过调控三线态激子浓度抑制淬灭，提高器件寿命。

【下阶段计划】电学行为分析——PH120电子迁移率减弱，工作电压升高~5%，电荷传输特性变化对电荷平衡/激子复合区影响通过阻抗谱、AREL+Setfos仿真评估



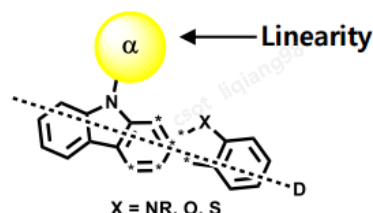
长寿命分析



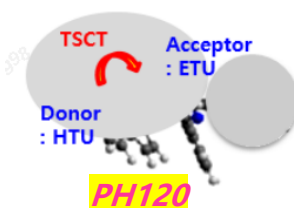
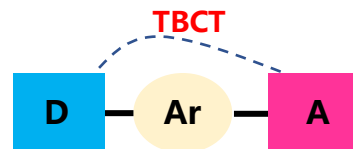
TADF型N-host 增加三线态激子
能量转移通道，减少浓度淬灭



P-host结构

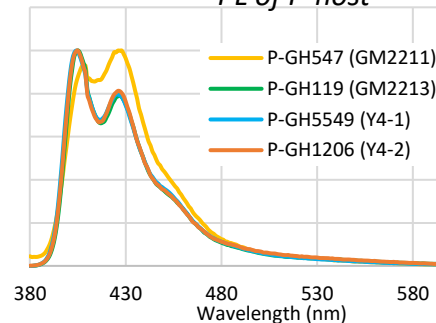


N-host结构

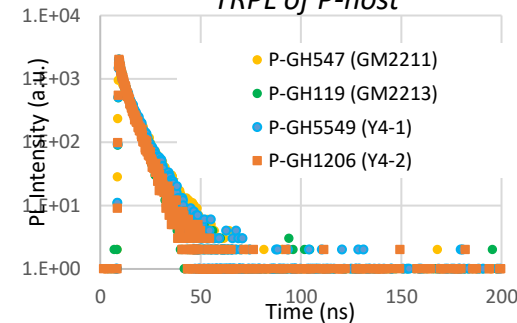


光物理性质分析

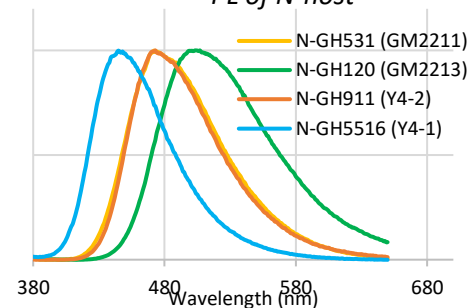
PL of P-host



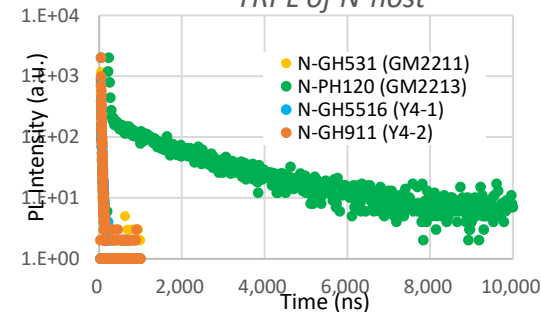
TRPL of P-host



PL of N-host



TRPL of N-host



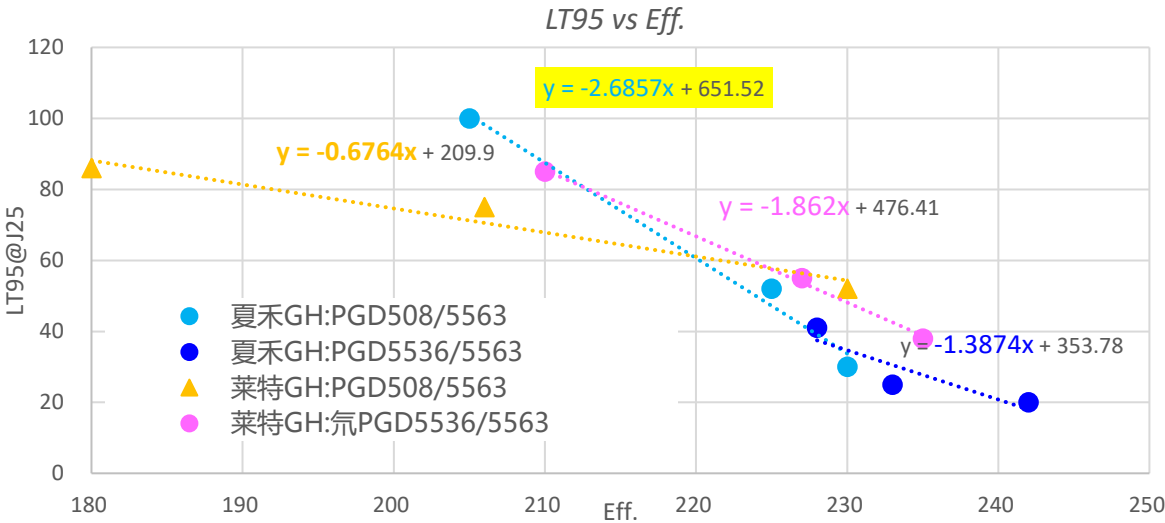
P-HOST	Peak (nm)	τ_1/τ_2 (ns/us)	N-HOST	Peak (nm)	τ_1/τ_2 (ns/us)
GH547	408/426	5.8/---	GH531	472	7.1/---
GH119	405/427	5.3/---	GH120	507	160ns/2.5us
GH5549	404/426	5.6/---	GH5516	445	5.6/---
GH1206	405/426	4.6/---	GH911	473	4.8/---

GH对寿命影响分析_电荷平衡影响

【GH电学影响】优化P/N-host比例，**寿命收益40%~90%**，**效率降低~7%-12%**，效率与寿命呈典型Trade-off关系

【电荷平衡分析进展】GH&PGD共同决定EML电平衡，基于P-GH5549/N-GH5516/SPGD508组合优化P/N比例，提高寿命的同时效率折损最小。

【下阶段计划】基于电荷注入/传输动力学分析厘清GH/PGD的关键影响因子，寻优长寿命组合方案。



P-GH5549/N-GH5516 SPGD508:SFGD5563 (20251225)	P/N ratio	Vd	CIE x	CIE y	C.E.(cd/A)	LT95@J25
	6/4	3.15	0.264	0.714	205	100
	5/5	3.10	0.260	0.720	225	52
	4/6	3.12	0.260	0.720	230	30
P-GH5549/N-GH5516 SPGD5536:SFGD5563 (20260227)	P/N ratio	Vd	CIE x	CIE y	C.E.(cd/A)	LT95@J25
	6/4	3.31	0.254	0.724	228	41
	5/5	3.13	0.256	0.722	233	25
	4/6	3.10	0.255	0.723	242	20
P-1206/N-911 SPGD508:SFGD5563 (20260310)	P/N ratio	Vd	CIE x	CIE y	C.E.(cd/A)	LT95@J25
	6/4	3.18	0.265	0.715	180	86
	5/5	3.20	0.264	0.716	206	75
	4/6	3.05	0.261	0.718	230	52
P-1206/N-911 氙SPGD5536:SFGD5563 (20260305)	P/N ratio	Vd	CIE x	CIE y	C.E.(cd/A)	LT95@J25
	6/4	3.28	0.229	0.743	210	85
	5/5	3.17	0.237	0.737	227	56
	4/6	3.13	0.234	0.739	235	38

高效率Y4方案

P/N比例调控下的寿命——效率Trade-off关系

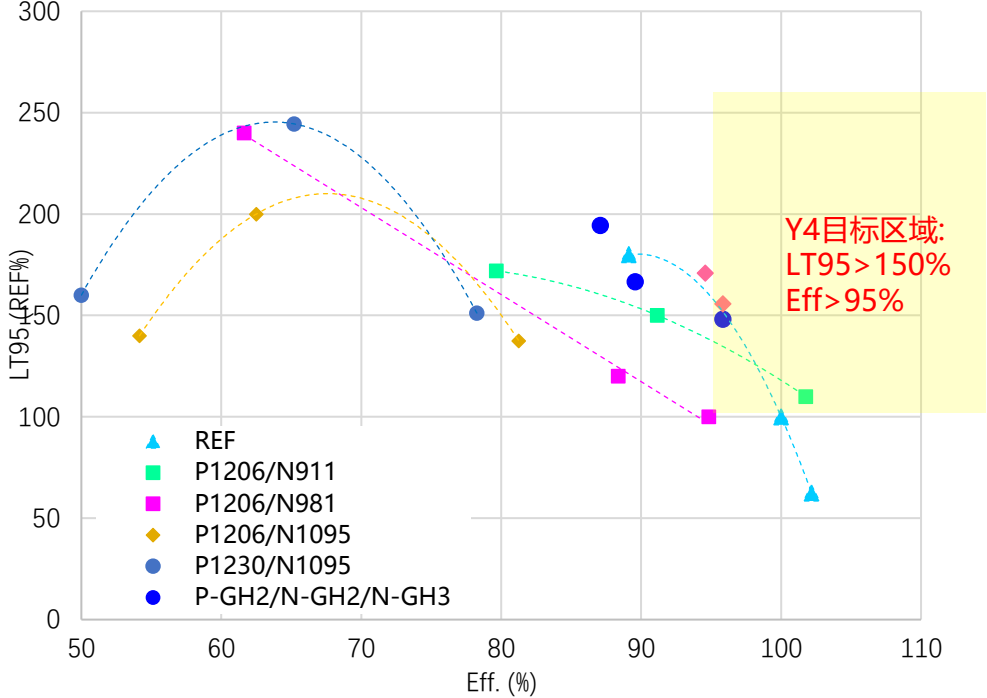
Y4双组分GH评估

序号	P-GH	N-GH	PGD	FGD	P/N比例	V@J10	CE	LT95
REF	P-GH2	N-GH2	PGD-3	FGD-2	6:4	102	89	208
					5:5	100	100	100
					4:6	98	102	63
1	1230	1095			6:4	95	50	160
					5:5	96	65	244
					4:6	98	78	151
2	1206	1095			6:4	102	54	140
					5:5	101	63	200
					4:6	100	81	138
3	1206	N911			6:4	96	80	172
					5:5	100	91	150
					4:6	96	102	110
4	1206	N981			6:4	100	62	240
					5:5	102	88	120
					4:6	98	95	100

基于电压关系评估迁移率特性：
空穴迁移率： P1230>P1206
电子迁移率： N911>N981>N1095

如何量化P/N-host的传输特性？
(HOD&EOD、阻抗谱)

不同Host P/N调控的效率-寿命 Trade-off关系



三组分GH共混 (Y4主方案)

序号	P-GH1	N-GH1	N-GH2	PGD	FGD	P/N比例	V@J10	CE	LT95
REF	P-GH2	N-GH2	/			5:5	100	100	100
1	P-GH2	N-GH2	N-GH3	PGD-3	FGD-2	5:(7:3)	103	96	148
						5:(6:4)	104	90	167
						5:(5:5)	103	87	194
2	1206	N911				4:(7:3)	96	96	156
						4:(5:5)	100	95	171

溶解度NG

Y4主方案